

The Hong Kong Polytechnic University

Subject Description Form

Subject Code	AAE5211
Subject Title	Spacecraft Dynamics
Credit Value	3
Level	5
Pre-requisite/ Co-requisite/ Exclusion	Nil
Objectives	To equip students with an understanding of theoretical and practical spacecraft dynamics from the perspective of dynamical systems, using a cohesive and integrated approach to mathematical modelling, analysis, and control of advanced space vehicles.
Intended Learning Outcomes	Upon completion of this subject, students will be able to: - Demonstrate a thorough understanding of the principles of spacecraft dynamics. - Apply theoretical concepts to analyse and solve practical spacecraft dynamics problems. - Design spacecraft trajectories, develop control algorithms, and assess spacecraft performance. - Recognise the importance of and exhibit the ability to engage in lifelong learning endeavours.
Subject Synopsis/ Indicative Syllabus	Introduction to Spacecraft Dynamics: Overview of spacecraft dynamics principles; Coordinate systems and transformations used in spacecraft dynamics; Basic concepts of rigid body dynamics and kinematics applicable to spacecraft. Spacecraft Kinematics and Dynamics: Position, velocity, and acceleration in space; Relative motion and reference frames; Forces and moments acting on spacecraft; Rotational kinematics; Dynamic differential equations of spacecraft; External torques on spacecraft. Orbit Determination and Propagation: Orbit determination techniques using ground-based and onboard sensors; Propagation of orbital elements and trajectory prediction. Attitude Dynamics and Control: Attitude representation and dynamics; Attitude control systems and stabilisation techniques. Guidance and Navigation: Principles of guidance and navigation in space; Orbit transfer manoeuvres and rendezvous operations.

	Spacecraft Formation Flying: Dynamics of multiple spacecraft formations; Control strategies for maintaining formation configurations; Mission Design and Analysis: Mission requirements and constraints; Trajectory optimisation and mission analysis techniques; Case Studies and Applications.					
Teaching/Learning Methodology	The teaching and learning methods include lectures, mini-project and homework assignments. Technical/practical examples and problems will be raised and discussed in class. Project is designed to evaluate the spacecraft dynamics.					
	Teaching/Learning Methodology	Outcomes				
		a	b	c	d	
		Lecture	✓	✓	✓	✓
		Mini-project	✓	✓	✓	✓
		Homework assignments	✓	✓	✓	✓
Assessment Methods in Alignment with Intended Learning Outcomes	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)			
			a	b	c	d
	1. Homework assignments	25%	✓	✓	✓	✓
	2. Mini-project	25%	✓	✓	✓	✓
	3. Final examination	50%	✓	✓	✓	
	Total	100%				
	Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes: 0.5 × Continuous Assessment + 0.5 × Final Examination Continuous assessment, comprising homework assignments and project report, functions to track students' advancement, facilitate self-evaluation against intended learning goals, and foster the amalgamation of knowledge gained from lectures. Conversely, the final examination objectively assesses students' knowledge acquisition, capacity to understand and critically analyse problems independently, and overall attainment of subject learning objectives.					
	Class contact:					
	▪ Lecture				39 Hrs.	

Student Study Effort Expected	Other student study effort:	
	▪ Self-study	78 Hrs.
	Total student study effort	117 Hrs.
Reading List and References	1. Spacecraft Dynamics and Control: The Embedded Model Control Approach, 1st Edition, 2018, E. Canuto <i>et al.</i> , Elsevier Ltd 2. Modelling and Simulation of Aerospace Vehicle Dynamics, 3rd Edition, 2014, P. Zipfel, AIAA 3. Spacecraft Dynamics and Control: An Introduction, 1st Edition, 2016, A. de Ruiter, C. Damaren, & J. Forbes, Wiley. 4. Space Vehicle Dynamics and Control, 2nd Edition, 2008, B. Wie, AIAA.	

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